Experiment No.	1
Experiment Title	Loan Approval Prediction [1]
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1 Objective

The objective of this experiment is to build an end-to-end supervised machine learning pipeline using K-Nearest Neighbors (KNN) to predict loan application approval (*Approved* or *Rejected*). The workflow covers data preprocessing, exploratory analysis, statistical feature selection, model training, and evaluation using classification metrics.

2 Problem Statement

Given an applicant's financial and demographic details (11 attributes including income, loan amount, CIBIL score, and assets), the task is to predict the binary target `loan_status` (*Approved* vs. *Rejected*) using a KNN classifier for automated loan pre-screening.

3 Dataset Description

Dataset Name	Loan Approval Dataset
Dataset Source	`loan_approval_dataset.csv`
Number of Samples	4269
Number of Features	11 (excluding `loan_id` and target `loan_status`)
Target Classes	2 (*Approved*: 2656, *Rejected*: 1613)
Missing Values	None (0 missing values)
Train-Test Split	80% train (3415) – 20% test (854), `random_state=42`

4 Exploratory Data Analysis

The dataset consists of 4269 rows without missing values or duplicates. The unique identifier `loan_id` was dropped. The target exhibits a moderate 62:38 class balance.

- Dataset overview – 4269 rows, 13 columns, no missing values, no duplicate rows.
- Statistical summary – mean CIBIL score $\approx$ 600, mean annual income $\approx$ \$58.65 lakh, loan terms ranging from 2 to 20 years.
- Missing value analysis – 0 missing values in every column.
- Class distribution – 2656 approved vs 1613 rejected applications, a mild class imbalance (roughly 62:38).
- Correlation matrix – computed over all numeric columns.

- Feature distribution – histograms of all nine numeric columns.

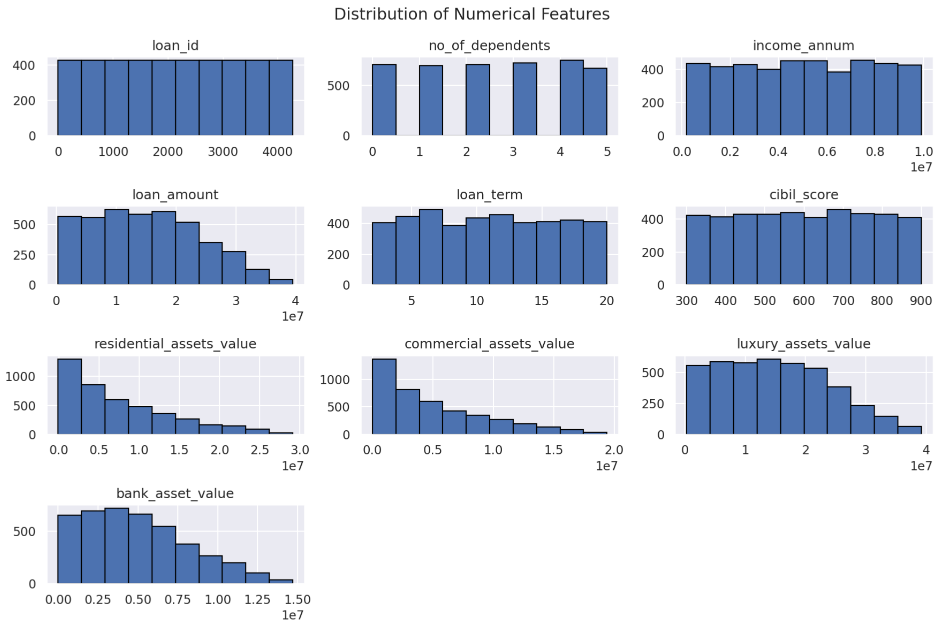

Figure 1: Distribution of numerical features.

Inference: Numerical features like income and assets show uniform distributions across their ranges. The CIBIL score ranges uniformly between 300 and 900, while loan term acts as an ordinal attribute.

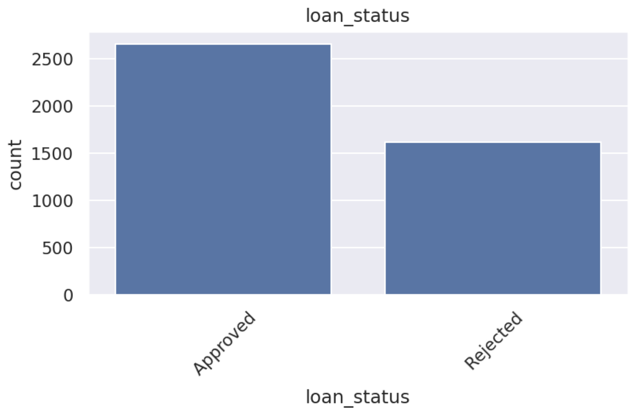

Figure 2: Class distribution of target variable `loan_status`.

Inference: The moderate class distribution (62% approved vs. 38% rejected) requires evaluating precision, recall, and F1-score alongside simple accuracy.

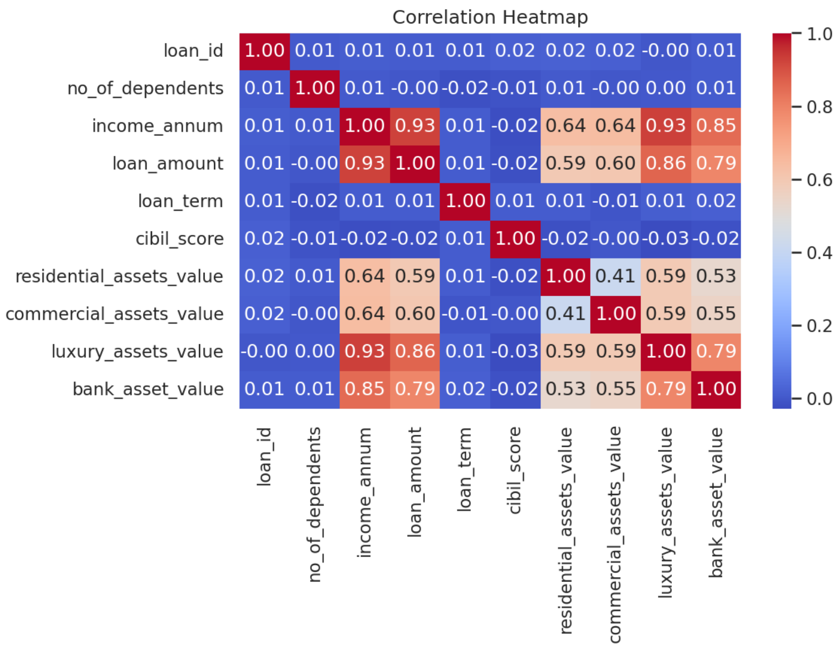

Figure 3: Correlation heatmap of numeric features.

Inference: Moderate correlation (0.4–0.6) exists among asset variables and between income and loan amount. CIBIL score shows negligible linear correlation with numerical features, indicating unique predictive signal.

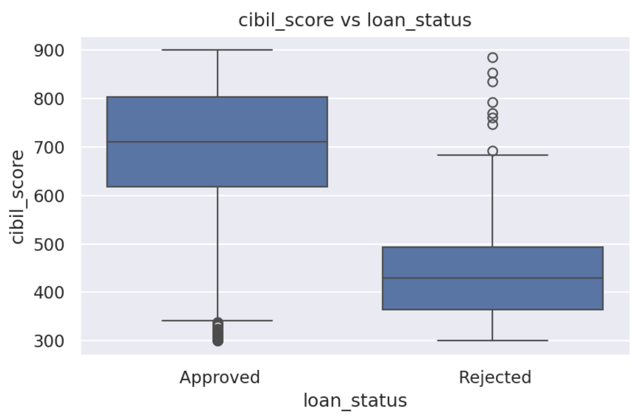

Figure 4: CIBIL score distribution across loan status classes.

Inference: Strong separation is observed; approved applications are concentrated above CIBIL score 550, while rejected ones fall below 550.

5 Data Preprocessing

- **Cleaning:** Verified zero missing values and removed duplicates. Dropped non-predictive `loan_id`.
- **Encoding:** Categorical features (`education`, `self_employed`) and target (`loan_status`) were encoded using `LabelEncoder`.

- **Scaling:** Applied `StandardScaler` to normalize continuous features for distance calculations.
- **Data Splitting:** Split dataset into 80% training and 20% testing sets (`random_state=42`).

6 Feature Selection

Chi-Square and ANOVA F-tests were applied to rank feature relevance.

Table 2: Top features ranked by statistical scores.

Feature	Chi-Square Score	ANOVA F-Score
cibil_score	418.64	6235.05
loan_term	11.09	55.23
no_of_dependents	0.32	1.40
loan_amount	0.16	1.11
luxury_assets_value	0.15	1.02
income_annum	0.16	0.98
residential_assets_value	0.17	0.88
commercial_assets_value	0.06	0.29
bank_asset_value	0.03	0.20
education	0.05	0.10
self_employed	0.0003	0.0005

Based on ANOVA F-scores, the top 5 features were selected: `no_of_dependents`, `loan_amount`, `loan_term`, `cibil_score`, and `luxury_assets_value`.

7 Machine Learning Algorithm: K-Nearest Neighbors

KNN is an instance-based algorithm that classifies a sample by assigning the majority label of its k-nearest neighbors based on distance metric calculations.

Distance Formula (Euclidean, $p = 2$):

$$d(x, x_i) = \sqrt{\sum_{j=1}^{n} (x_j - x_{ij})^2}$$

Decision Rule:

$$\hat{y} = \underset{i \in N_k(x)}{\text{mode}} \; y_i$$

8 Experimental Setup & Hyperparameters

Experiments were executed in standard Python 3.12 / scikit-learn environments using the following parameters:

- **n_neighbors (k):** 5 (Default)
- **Weights:** Uniform — **Metric:** Minkowski ($p = 2$, Euclidean)

9 Results

Table 3: KNN Model Performance Metrics.

Metric	Value
Accuracy	0.9297
Precision	0.9006
Recall	0.9119
F1-score	0.9063
ROC-AUC	0.9786

On 854 test samples, the classifier achieved 504 True Negatives, 32 False Positives, 28 False Negatives, and 290 True Positives.

10 Result Visualization

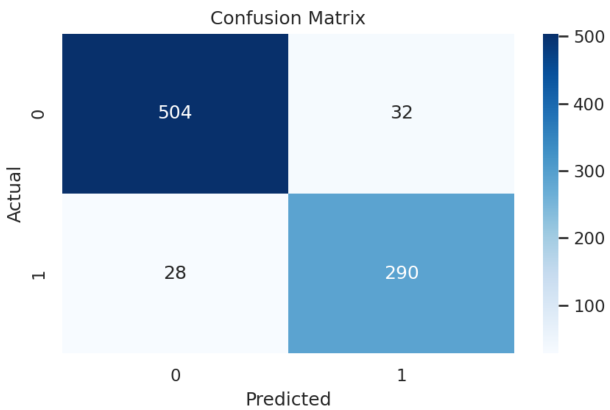

Figure 5: Confusion matrix on the test set.

Inference: Strong diagonal dominance confirms high accuracy across both classes with balanced error distribution.

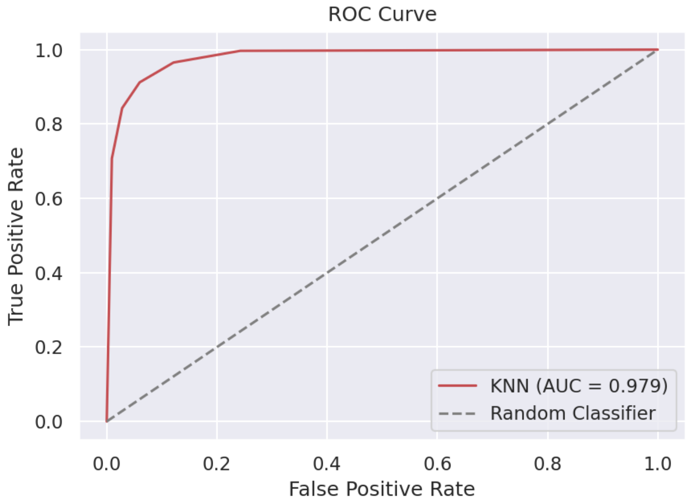

Figure 6: ROC curve for KNN classifier.

Inference: AUC of 0.979 shows excellent class separation across decision thresholds.

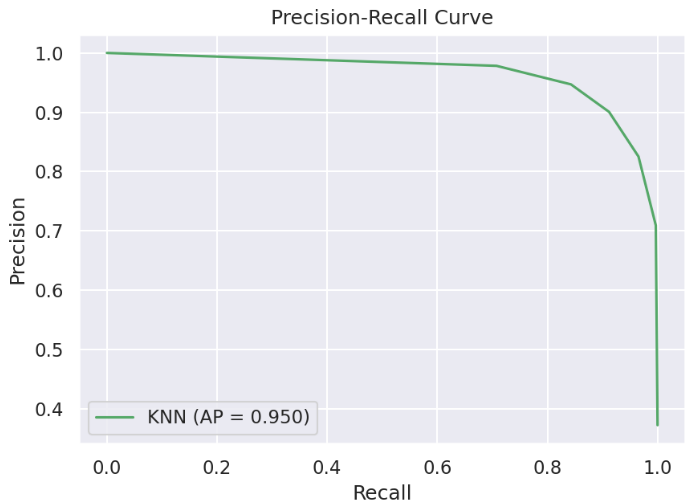

Figure 7: Precision-Recall curve.

Inference: High precision is maintained across all recall levels for the minority class.

11 Discussion

The baseline KNN model achieves strong results ($\approx$ 93% accuracy), largely driven by the high predictive power of `cibil_score`. Hyperparameter tuning over values of k and cross-validation could optimize performance further. Scaling optimizations would be needed for larger deployment datasets.

12 Conclusion

This study implemented a complete supervised learning workflow to address automated loan prescreening using K-Nearest Neighbors. Data preprocessing—including standard feature scaling and label encoding—ensured optimal distance computations for the KNN algorithm. Feature selection via Chi-Square and ANOVA F-tests revealed that creditworthiness (`cibil_score`) and loan duration (`loan_term`) dominate the predictive decision over raw asset values.

Evaluating the baseline model ($k = 5$) on an independent 20% test split yielded robust overall classification capabilities:

- **Accuracy:** 92.97%
- **Precision / Recall:** 90.06% / 91.19%
- **ROC-AUC:** 0.9786

Overall, the experiment highlights that rigorous preprocessing paired with feature selection enables simple non-parametric algorithms like KNN to achieve high accuracy and strong class separation in financial risk modeling tasks.

13 References

1. Chitraju Vishnu Vineeth, "Loan Approval ML Workflow," GitHub repository, 2026. [Online]. Available: https://github.com/VishnuVineeth14/ML_LAB_ASSIGNMENTS/blob/main/Assignment1/Loan_Approval_ML_Workflow.ipynb